

Complex Analysis solution

4.5 #6

Proof. By assumption, there is some $r > 0$ such that $f(z) = (z - z_0)^k g(z)$ on $D_r(z_0)$ where $g(z)$ is analytic and $g(z_0) \neq 0$. Since $g(z)$ is continuous, there is some $0 < r_1 < r$ such that $|g(z)| \neq 0$ on $D_{r_1}(z_0) \setminus \{z_0\}$. This implies $|f(z)| = |z - z_0|^k |g(z)| > 0$ on $D_{r_1}(z_0) \setminus \{z_0\}$.

Let $V = D_{r_2}(z_0)$ for some $0 < r_2 < r_1$. Then $f(z) \neq 0$ on $\bar{V} \setminus \{z_0\}$. So by construction, $f(z)$ has only one zero of order k at z_0 on V . That is, $f(z)$ has exactly k roots on V .

Since $f(z)$ continuous on the compact set ∂V , there is some $M > 0$ such that $|f(z)| \geq M$ on ∂V . Let $W = \{w : 0 < |w| < M\}$. We show $f(z) - w$ and $f(z)$ have the same number of zeroes on V .

Clearly $f(z) - w$ is analytic on V . Then note $f(z) - w$ has no zeroes on ∂V because

$$|f(z) - w| \geq ||f(z)| - |w|| \geq M - |w| > 0.$$

Finally we note $|(f(z) - w) - f(z)| = |w| < M \leq |f(z)|$ on V . By Rouché's theorem, $f(z) - w$ and $f(z)$ have the same number of zeroes on V . But we have showed $f(z)$ has exactly k roots on V . $\square$